Pennstander fonts (*experimental*) User's Guide for LaTeX

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1 What is Pennstander ?

Pennstander is a set of OpenType text and math fonts developed by Julius Ross, see github.com/juliusross1 for more information.

The text and the math fonts are licensed under the SIL Open Font License, Version 1.1.

They require LuaTeX or XeTeX as engine and the unicode-math package[1], if math fonts are required or just the fontspec package[2] otherwise.

2 Usage

2.1 Loading text fonts and math fonts

Several weights are available:

Thin / ExtraLight / Light / Regular / Medium / SemiBold / Bold

for text or math version.

Files Pennstander(math)-<weight>.fontspec are provided to ensure that Italic, Bold, BoldItalic ans (fake)Slanted variants are properly loaded.

A basic call for Pennstander text and math fonts with scaling could be:

```
\usepackage{pennstander-otf}
```

[1]Please read the documentation unicode-math.pdf.
[2]Please read the documentation fontspec.pdf.

It loads:

- `Pennstander-Regular` as main font;
- `PennstanderMath-Regular` as math font.

It defines `\eff` for *florin math* f.

Theorem 1 (Residue Theorem). Let f be analytic in the region G except for the isolated singularities $a_1, a_2, \dots, a_m$. If γ is a closed rectifiable curve in G which does not pass through any of the points a_k and if $\gamma \approx 0$ in G then

$$\frac{1}{2\pi i}\int_\gamma f = \sum_{k=1}^{m} n(\gamma; a_k)\mathrm{Res}(f; a_k).$$

Theorem 2 (Maximum Modulus). *Let G be a bounded open set in $\mathbb{C}$ and suppose that f is a continuous function on G^- which is analytic in G. Then*

$$\max\{|f(z)| : z \in G^-\} = \max\{|f(z)| : z \in \partial G\}.$$

2.2 Options

Options can be given within the package:

- `Weight=...`, within `Thin/ExtraLight/Light/Regular/Medium/SemiBold/Bold`;
- the boolean `no-text` fot not loading pennstander as main font;
- the boolean `no-math` fot not loading pennstander as math font;
- `math-style=...` for unicode-math option;
- `bold-style=...` for unicode-math option;
- `StylisticSet=...`
- `RawFeatureT=...` for text version;
- `RawFeatureM=...` for math version;
- `Scale=...`

2.3 Manual loading

It's also possible to load text and/or math fonts with `\setmainfont` and/or `\setmathfont`, thanks to `fontspec` files for example.

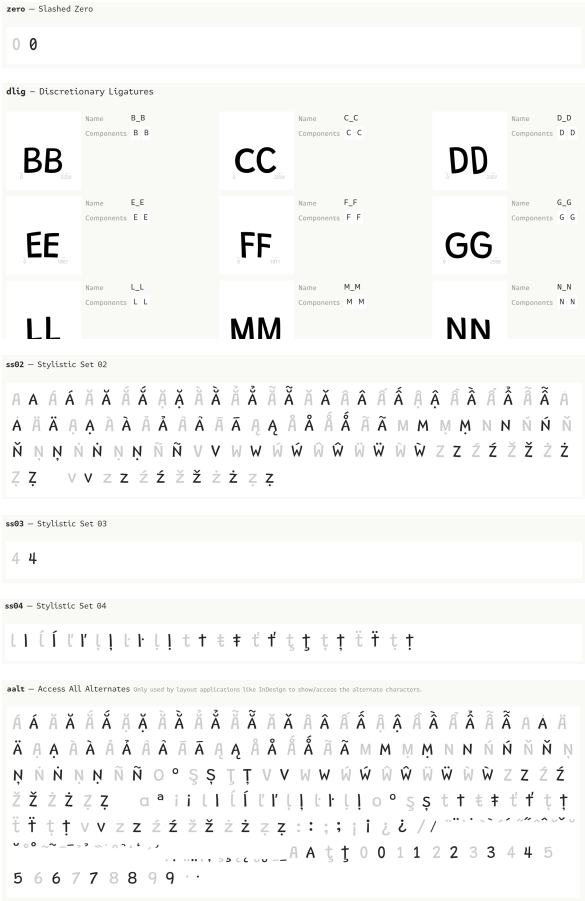
```
%w/o package loaded
\usepackage{fontspec}
\usepackage{unicode-math}
\setmainfont{Pennstander-<weight>}[options]
\setmainfont{PennstanderMath-<weight>}[options]
```

3 Openfeatures

3.1 For text font

Extra openfeatures for text font available are (screens are from fontdrop website):

- `aalt`: Access All Alternatives
- `ccmp`: Glyph Composition/Decomposition
- `locl`: Localized Forms
- `sups`: Superscript
- `numr`: Numerators
- `dnom`: Denominators
- `frac`: Fractions
- `ordn`: Ordinals
- `pnum`: Proportional Figures
- `tnum`: Tabular Figures
- `case`: Case-Sensitive Forms
- `dlig`: Discretionary Ligatures
- `liga`: Standard Ligatures
- `zero`: Slashed Zero
- `cpsp`: Capital Spacing
- `ss02`: Alternate A,M,N,Q,V,W,Z,v,z
- `ss03`: Alternate 4
- `ss04`: Alternate l,t
- `ss06`: Alternate ampersand
- `ss07`: Alternate 7



3.2 For math font

Extra openfeatures for math font available are:

- `aalt`: Access All Alternatives
- `dtls`: Dotless Forms
- `zero`: Slashed Zero
- `frac`: Fractions
- `ss01`: Alternate a,g,I,Iota
- `ss02`: Alternate A,M,N,Q,V,W,Z,v,z,Alpha,Mu,Rho,Zeta
- `ss03`: Alternate 4
- `ss04`: Alternate l,t
- `ss05`: Simplified Fraktur A,K,L,X,Y,Z,x,y
- `ss07`: Alternate 7

3.3 Unicode Private Use Area

Glyphs assigned to Unicode Private Use Area:

- Double-struck uppercase Greek letters: U+E1002–U+E1008
- Fraction Rule: U+E000
- Radical Rule: U+E001

4 List of glyphs

4.1 Text version

Table 1: Pennstander-Regular

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Basic Latin																
U+0000–000F		–	–	–	–	–	–	–	–	–	–	–	–		–	–
U+0020–002F		!	”	#	$	%	&	’	(	)	*	+	,	-	.	/
U+0030–003F	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
U+0040–004F	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
U+0050–005F	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
U+0060–006F	‘	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
U+0070–007F	p	q	r	s	t	u	v	w	x	y	z	{	|	}	~	–
Latin-1 Supplement																
U+00A0–00AF		¡	¢	£	¤	¥	¦	§	¨	©	ª	«	¬	-	®	¯
U+00B0–00BF	°	±	²	³	´	µ	¶	·	¸	¹	º	»	¼	½	¾	¿
U+00C0–00CF	À	Á	Â	Ã	Ä	Å	Æ	Ç	È	É	Ê	Ë	Ì	Í	Î	Ï
U+00D0–00DF	Ð	Ñ	Ò	Ó	Ô	Õ	Ö	×	Ø	Ù	Ú	Û	Ü	Ý	Þ	ß
U+00E0–00EF	à	á	â	ã	ä	å	æ	ç	è	é	ê	ë	ì	í	î	ï
U+00F0–00FF	ð	ñ	ò	ó	ô	õ	ö	÷	ø	ù	ú	û	ü	ý	þ	ÿ
Latin Extended-A																
U+0100–010F	Ā	ā	Ă	ă	Ą	ą	Ć	ć	Ĉ	ĉ	Ċ	ċ	Č	č	Ď	ď
U+0110–011F	Đ	đ	Ē	ē	Ĕ	ĕ	Ė	ė	Ę	ę	Ě	ě	Ĝ	ĝ	Ğ	ğ
U+0120–012F	Ġ	ġ	Ģ	ġ	Ĥ	ĥ	Ħ	ħ	–	ĩ	Ī	ī	Ĭ	ĭ	Ī	ī
U+0130–013F	İ	ı	–	–	Ĵ	ĵ	Ķ	ķ	ĸ	Ĺ	ĺ	Ļ	ļ	Ľ	ľ	Ŀ
U+0140–014F	ŀ	Ł	ł	Ń	ń	Ņ	ņ	Ň	ň	–	Ŋ	ŋ	Ō	ō	Ŏ	ŏ
U+0150–015F	Ő	ő	Œ	œ	Ŕ	ŕ	Ŗ	ŗ	Ř	ř	Ś	ś	Ŝ	ŝ	Ş	ş
U+0160–016F	Š	š	Ţ	ţ	Ť	ť	Ŧ	ŧ	Ũ	ũ	Ū	ū	Ŭ	ŭ	Ů	ů

Table 1: Pennstander-Regular *cont.*

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
U+0170–017F	Ű	ű	Ų	ų	Ŵ	ŵ	Ŷ	ŷ	Ÿ	Ź	ź	Ż	ż	Ž	ž	–
Latin Extended-B																
U+0180–018F	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	ə
U+0190–019F	–	–	ƒ	–	–	–	–	–	–	–	–	–	–	–	–	–
U+01A0–01AF	Ơ	ơ	–	–	–	–	–	–	–	–	–	–	–	–	–	Ư
U+01B0–01BF	ư	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
U+01C0–01CF	–	–	–	–	DŽ	Dž	dž	LJ	Lj	lj	NJ	Nj	nj	Ǎ	ǎ	Ǐ
U+01D0–01DF	ǐ	Ǒ	ǒ	Ǔ	ǔ	Ǖ	ǖ	Ǘ	ǘ	Ǚ	ǚ	Ǜ	ǜ	–	–	–
U+01E0–01EF	–	–	–	–	–	–	Ǧ	ǧ	–	–	Ǫ	ǫ	–	–	–	–
U+01F0–01FF	–	–	–	–	–	–	–	–	–	–	Ǻ	ǻ	Ǽ	ǽ	Ǿ	ǿ
U+0200–020F	Ȁ	ȁ	Ȃ	ȃ	Ȅ	ȅ	Ȇ	ȇ	Ȉ	ȉ	Ȋ	ȋ	Ȍ	ȍ	Ȏ	ȏ
U+0210–021F	Ȑ	ȑ	Ȓ	ȓ	Ȕ	ȕ	Ȗ	ȗ	Ș	ș	Ț	ț	–	–	–	–
U+0220–022F	–	–	–	–	–	–	–	–	–	–	Ȟ	ȟ	Ƞ	ȠȠ	–	–
U+0230–023F	Ȱ	ȱ	Ȳ	ȳ	–	–	–	ɟ	–	–	–	–	–	–	–	–
IPA Extensions																
U+0250–025F	–	–	–	–	–	–	–	–	–	ə	–	–	–	–	–	–
Spacing Modifier Letters																
U+02B0–02BF	–	–	–	–	–	–	–	–	–	–	–	–	ʼ	–	–	–
U+02C0–02CF	–	–	–	–	–	–	ˆ	ˇ	–	ˉ	–	–	–	–	–	–
U+02D0–02DF	–	–	–	–	–	–	–	–	˘	˙	˚	˛	˜	˝	–	–
Combining Diacritical Marks																
U+0300–030F	̀	́	̂	̃	̄	̅	̆	̇	̈	̉	̊	̋	̌	̍	̎	̏
U+0310–031F	–	̐	̑	–	–	–	–	–	–	–	–	̕	–	–	–	–
U+0320–032F	–	–	–	̣	̤	–	̦	̧	̨	–	–	–	–	–	̪	–
U+0330–033F	–	̳	–	–	–	–	–	̷	̸	–	–	–	–	–	–	–
Latin Extended Additional																
U+1E00–1E0F	–	–	–	–	–	–	–	–	Ḉ	ḉ	–	–	Ḍ	ḍ	Ḏ	ḏ
U+1E10–1E1F	–	–	–	–	Ḕ	ḕ	Ḗ	ḗ	–	–	–	–	Ḝ	ḝ	–	–
U+1E20–1E2F	Ḡ	ḡ	–	–	Ḥ	ḥ	–	–	–	–	Ḫ	ḫ	–	–	Ḭ	ḭ
U+1E30–1E3F	–	–	–	–	–	–	Ḷ	ḷ	–	–	Ḻ	ḻ	–	–	–	–
U+1E40–1E4F	–	–	Ṃ	ṃ	Ṅ	ṅ	Ṇ	ṇ	Ṉ	ṉ	–	–	Ṑ	ṑ	Ṓ	ṓ
U+1E50–1E5F	Ṕ	ṕ	Ṗ	ṗ	–	–	–	–	–	–	Ṛ	ṛ	–	–	Ṝ	ṝ
U+1E60–1E6F	Ṡ	ṡ	Ṣ	ṣ	Ṥ	ṥ	Ṧ	ṧ	Ṩ	ṩ	–	–	Ṭ	ṭ	Ṯ	ṯ
U+1E70–1E7F	–	–	–	–	–	–	–	–	Ṵ	ṵ	Ṷ	ṶṶ	–	–	–	–
U+1E80–1E8F	Ẁ	ẁ	Ẃ	ẃ	Ẅ	ẅ	–	–	–	–	–	–	–	–	Ẏ	ẏ

Table 1: Pennstander-Regular *cont.*

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
U+1E90–1E9F	–	–	Ẓ	ẓ	–	–	–	ẗ	–	–	–	–	–	–	ẞ	–
U+1EA0–1EAF	Ạ	ạ	Ả	ả	Ấ	ấ	Ầ	ầ	Ẩ	ẩ	Ẫ	ẫ	Ậ	ậ	Ắ	ắ
U+1EB0–1EBF	Ằ	ằ	Ẳ	ẳ	Ẵ	ẵ	Ặ	ặ	Ẹ	ẹ	Ẻ	ẻ	Ẽ	ẽ	Ế	ế
U+1EC0–1ECF	Ề	ề	Ể	ể	Ễ	ễ	Ệ	ệ	Ỉ	ỉ	Ị	ị	Ọ	ọ	Ỏ	ỏ
U+1ED0–1EDF	Ố	ố	Ồ	ồ	Ổ	ổ	Ỗ	ỗ	Ộ	ộ	Ớ	ớ	Ờ	ờ	Ở	ở
U+1EE0–1EEF	Ỡ	ỡ	Ợ	ợ	Ụ	ụ	Ủ	ủ	Ứ	ứ	Ừ	ừ	Ử	ử	Ữ	ữ
U+1EF0–1EFF	Ự	ự	Ỳ	ỳ	Ỵ	ỵ	Ỷ	ỷ	Ỹ	ỹ	–	–	–	–	–	–

General Punctuation

U+2000–200F				–	–	–	–	–	–			–	–	–	–	–
U+2010–201F	–	–	–	‐	―	–	–	–	‘	’	‚	–	“	”	„	–
U+2020–202F	†	‡	•	–	–	–	…	–	–	–	–	–	–	–	–	
U+2030–203F	‰	–	–	–	–	–	–	–	–	‹	›	–	–	–	–	–
U+2040–204F	–	–	–	–	⁄	–	–	–	–	–	–	–	–	–	–	–

Superscripts and Subscripts

U+2070–207F	⁰	–	–	–	⁴	⁵	⁶	⁷	⁸	⁹	–	–	–	–	–	–

Currency Symbols

U+20A0–20AF	–	₠	–	₣	₤	–	₦	₱	–	₩	–	₫	€	₭	–	–
U+20B0–20BF	–	₰	₲	–	–	₵	–	–	–	₹	₺	–	₼	₽	–	–

Letterlike Symbols

U+2110–211F	–	–	–	–	–	–	№	–	–	–	–	–	–	–	–	–
U+2120–212F	–	–	™	–	–	–	–	–	–	–	–	–	–	–	–	–

Number Forms

U+2150–215F	–	–	–	–	–	–	–	–	–	–	–	⅛	⅜	⅝	⅞	–

Mathematical Operators

U+2200–220F	–	–	–	–	–	∅	–	–	–	–	–	–	–	–	□	–
U+2210–221F	–	–	−	–	–	∕	–	–	–	∙	–	–	–	–	–	–
U+2240–224F	–	–	–	–	–	–	–	–	≈	–	–	–	–	–	–	–
U+2260–226F	≠	–	–	–	≤	≥	–	–	–	–	–	–	–	–	–	–

Geometric Shapes

U+25C0–25CF	–	–	–	–	–	–	–	–	–	–	◊	–	–	–	–	–

Miscellaneous Shapes

U+2660–266F	♠	♡	♢	♣	♤	♥	♦	♧	–	♩	♪	♫	♬	–	–	–

Table 1: Pennstander-Regular *cont.*

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Alphabetic Presentation Forms																
U+FB00–FB0F	–	ﬁ	ﬂ	–	–	–	–	–	–	–	–	–	–	–	–	–

Total number of glyphs shown from Pennstander-Regular: 671

4.2 Math version

Table 2: PennstanderMath-Regular

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Basic Latin																
U+0000–000F	–	–	–	–	–	–	–	–	–	–	–	–	–		–	–
U+0020–002F		!	"	#	–	%	&	'	(	)	*	+	,	-	.	/
U+0030–003F	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
U+0040–004F	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
U+0050–005F	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
U+0060–006F	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
U+0070–007F	p	q	r	s	t	u	v	w	x	y	z	{	|	}	~	–
Latin-1 Supplement																
U+00A0–00AF		¡	–	–	–	–	¦	§	¨	©	ª	«	¬	-	®	¯
U+00B0–00BF	°	±	²	³	´	µ	¶	·	¸	¹	º	»	–	–	–	¿
U+00D0–00DF	–	–	–	–	–	–	–	×	–	–	–	–	–	–	–	–
U+00F0–00FF	ð	–	–	–	–	–	–	÷	–	–	–	–	–	–	–	–
Latin Extended-A																
U+0130–013F	–	ı	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Latin Extended-B																
U+0180–018F	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	ə
U+0190–019F	–	–	ƒ	–	–	–	–	–	–	–	–	–	–	–	–	–
U+0230–023F	–	–	–	–	–	–	–	ȷ	–	–	–	–	–	–	–	–
IPA Extensions																
U+0250–025F	–	–	–	–	–	–	–	–	–	ə	–	–	–	–	–	–
Spacing Modifier Letters																
U+02B0–02BF	–	–	–	–	–	–	–	–	–	–	–	–	ʼ	–	–	–

Table 2: PennstanderMath-Regular *cont.*

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
U+02C0–02CF	–	–	–	–	–	–	ˆ	ˇ	–	ˉ	–	–	–	–	–	–
U+02D0–02DF	–	–	–	–	–	–	–	–	˘	˙	˚	˛	˜	˝	–	–

Combining Diacritical Marks

U+0300–030F	̀	́	̂	̃	̄	̅	̆	̇	̈	̉	̊	̋	̌	̍	̎	̏
U+0310–031F	̐	̑	̒	–	–	–	–	–	–	–	–	̕	–	–	–	–
U+0320–032F	–	–	–	̠	̡	–	̢	̣	̤	–	–	–	–	–	–	–
U+0330–033F	–	̳	̴	–	–	–	–	̸	̷	–	–	–	–	–	–	–
U+0340–034F	–	–	–	–	–	–	–	–	–	–	–	–	–	͡	–	–

Greek and Coptic

U+0390–039F	–	Α	Β	Γ	Δ	Ε	Ζ	Η	Θ	Ι	Κ	Λ	Μ	Ν	Ξ	Ο
U+03A0–03AF	Π	Ρ	–	Σ	Τ	Υ	Φ	Χ	Ψ	Ω	–	–	–	–	–	–
U+03B0–03BF	–	α	β	γ	δ	ε	ζ	η	θ	ι	κ	λ	μ	ν	ξ	ο
U+03C0–03CF	π	ρ	ς	σ	τ	υ	φ	χ	ψ	ω	–	–	–	–	–	–
U+03D0–03DF	–	ϑ	–	–	–	ϕ	ϖ	–	–	–	–	–	–	–	–	–
U+03F0–03FF	ϰ	ϱ	–	–	ϴ	ϵ	϶	–	–	–	–	–	–	–	–	–

Phonetic Extensions Supplement

U+1DA0–1DAF	–	–	–	–		–	–	–	–	–	–	–	–	–	–	–

General Punctuation

U+2000–200F				–	–	–	–	–	–			–	–	–	–	–
U+2010–201F	–	–	–	‐	‒	–	‖	‗	‘	’	‚	–	“	”	„	–
U+2020–202F	†	‡	•	–	–	–	…	–	–	–	–	–	–	–	–	
U+2030–203F	‰	–	′	″	‴	‵	‶	‷	–	‹	›	–	–	–	‾	–
U+2040–204F	–	–	–	–	⁄	–	–	–	–	–	–	–	–	–	–	–
U+2050–205F	–	–	–	–	–	–	–	⁗	–	–	–	–	–	–	–	–

Superscripts and Subscripts

U+2070–207F	⁰	–	–	–	⁴	⁵	⁶	⁷	⁸	⁹	–	–	–	–	–	–

Combining Diacritical Marks for Symbols

U+20D0–20DF	⃐	⃑	–	–	–	–	⃖	⃗	–	–	–	⃛	⃜	–	–	–
U+20E0–20EF	–	⃡	–	–	–	–	–	–	⃤	–	–	–	⃥	⃦	⃧	⃨
U+20F0–20FF	⃩	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

Letterlike Symbols

U+2100–210F	–	–	ℂ	–	–	–	–	–	–	–	–	ℋ	ℌ	ℍ	ℎ	ℏ
U+2110–211F	ℐ	ℑ	ℒ	ℓ	–	ℕ	№	–		ℙ	ℚ	ℛ	ℜ	ℝ	–	–
U+2120–212F	–	–	™	–	ℤ	–	–	℧	ℨ	–	–	Å	ℬ	ℭ	–	–
U+2130–213F	ℰ	ℱ	–	ℳ	–					–	–	–	–	–	ℾ	ℿ

Table 2: PennstanderMath-Regular *cont.*

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
U+2140–214F	⅀	–	–	–	–	ⅅ	–	–			–	–	–	–	–	–
Arrows																
U+2190–219F	←	↑	→	↓	↔	↕	↖	↗	↘	↙	↚	↛	–	–	↞	↟
U+21A0–21AF	↠	↡	↢	↣	↤	↥	↦	↧	↨	↩	↪	↫	↬	–	↮	–
U+21B0–21BF	–	–	–	–	–	–	–	–	–	–	–	–	↼	↽	↾	↿
U+21C0–21CF	⇀	⇁	⇂	⇃	⇄	⇅	⇆	⇇	⇈	⇉	⇊	⇋	⇌	⇍	⇎	⇏
U+21D0–21DF	⇐	⇑	⇒	⇓	⇔	⇕	⇖	⇗	⇘	⇙	–	–	–	–	–	–
U+21E0–21EF	⇠	⇡	⇢	⇣	⇤	⇥	–	–	–	–	–	–	–	–	–	–
U+21F0–21FF	–	–	–	–	–	⇵	–	⇦	⇨	⇔	⇍	⇏	⇎	–	–	–
Mathematical Operators																
U+2200–220F	∀	∁	∂	∃	∄	∅	–	∇	∈	∉	–	∋	∌	–	∎	∏
U+2210–221F	∐	∑	–	∓	∔	∕	∖	∗	∘	∙	√	∛	∜	∝	∞	∟
U+2220–222F	∠	∡	∢	∣	∤	∥	∦	∧	∨	∩	∪	∫	∬	∭	∮	∯
U+2230–223F	∰	∱	∲	∳	∴	∵	∶	∷	∸	∹	∺	–	∼	∽	∾	∿
U+2240–224F	≀	≁	≂	≃	–	≅	≆	–	≈	≉	–	–	≌	≍	–	–
U+2250–225F	≐	≑	≒	≓	≔	≕	≖	≗	≘	≙	≚	≛	≜	≝	≞	≟
U+2260–226F	≠	≡	≢	≣	≤	≥	≦	≧	≨	≩	≪	≫	≬	≭	≮	≯
U+2270–227F	≰	≱	≲	≳	≴	≵	≶	≷	≸	≹	≺	≻	≼	≽	≾	≿
U+2280–228F	⊀	⊁	⊂	⊃	⊄	⊅	⊆	⊇	⊈	⊉	⊊	⊋	⊌	⊍	⊎	⊏
U+2290–229F	⊐	⊑	⊒	⊓	⊔	⊕	⊖	⊗	⊘	⊙	⊚	⊛	⊜	⊝	⊞	⊟
U+22A0–22AF	⊠	⊡	⊢	⊣	⊤	⊥	⊦	⊧	⊨	⊩	⊪	⊫	⊬	⊭	⊮	⊯
U+22B0–22BF	⊰	⊱	⊲	⊳	⊴	⊵	⊶	⊷	–	⊹	⊺	⊻	⊼	⊽	–	⊿
U+22C0–22CF	⋀	⋁	⋂	⋃	⋄	⋅	⋆	⋇	⋈	⋉	⋊	⋋	⋌	≍	⋎	⋏
U+22D0–22DF	⋐	⋑	⋒	⋓	⋔	–	⋖	⋗	⋘	⋙	⋚	⋛	⋜	⋝	⋞	⋟
U+22E0–22EF	⋠	⋡	⋢	⋣	⋤	⋥	⋦	⋧	⋨	⋩	⋪	⋫	⋬	⋭	⋮	⋯
U+22F0–22FF	⋰	⋱	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Miscellaneous Technical																
U+2300–230F	–	–	–	–	–	⍱	⍲	–	⌈	⌉	⌊	⌋	–	–	–	–
U+2310–231F	⌐	–	–	–	–	–	–	–	–	⌙	–	–	⌜	⌝	⌞	⌟
U+2320–232F	⌠	⌡	–	–	–	–	–	–	–	–	–	–	–	–	–	–
U+23A0–23AF	–	⎡	⎢	⎣	⎤	⎥	⎦	⎧	⎨	⎩	⎫	–	–	–	⎭	⎯
U+23B0–23BF	–	–	–	–	⎴	⎵	–	⎷	–	–	–	–	–	–	–	–
U+23D0–23DF	–	–	–	–	–	–	–	–	–	–	–	–	⏜	⏝	⏞	⏟
Geometric Shapes																
U+25A0–25AF	–	□	–	–	–	–	–	–	–	–	–	–	–	–	–	–
U+25C0–25CF	–	–	–	–	–	–	–	–	–	–	◊	–	–	–	–	–

Table 2: PennstanderMath-Regular *cont.*

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Miscellaneous Shapes																
U+2660–266F	♠	♡	♢	♣	♤	♥	♦	♧	–	♩	♪	♫	♬	♭	♮	♯
Dingbats																
U+2710–271F	–	–	–	✓	–	–	–	–	–	–	–	–	–	–	–	–
Miscellaneous Mathematical Symbols-A																
U+27C0–27CF	–	–	⊥	–	–	–	–	–	–	–	–	–	–	–	–	–
U+27D0–27DF	–	–	–	–	–	–	–	–	⊥	⊤	–	–	–	–	–	–
U+27E0–27EF	–	–	–	–	–	–	⟦	⟧	⟨	⟩	⟪	⟫	–	–	⦇	⦈
Supplemental Arrows-A																
U+27F0–27FF	–	–	–	–	–	⟵	⟶	⟷	⟸	⟹	⟺	⟼	⟾	⟿	⟽	–
Supplemental Arrows-B																
U+2900–290F	–	–	–	–	–	–	⤌	⤍	–	–	–	–	–	–	–	–
U+2920–292F	–	–	–	⤡	⤢	⤣	⤤	–	–	–	–	–	–	–	–	–
Miscellaneous Mathematical Symbols-B																
U+2980–298F	⦚	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
U+29B0–29BF	–	–	–	–	–	⦵	⦶	⦷	⦸	⦹	⦺	⦻	⦼	⦽	⦾	⦿
U+29C0–29CF	⧀	⧁	–	–	⧄	⧅	⧆	⧇	⧈	–	–	–	–	–	–	–
U+29E0–29EF	–	–	⧧	–	–	–	–	–	–	–	–	–	–	–	–	–
Supplemental Mathematical Operators																
U+2A00–2A0F	⨀	⨁	⨂	⨃	⨄	⨅	⨆	–	–	⨯	⨊	–	⨌	⨍	–	⨏
U+2A10–2A1F	–	⨐	–	–	–	–	–	–	–	–	–	–	–	–	–	–
U+2A20–2A2F	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	⨯
U+2A30–2A3F	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	⨿
U+2A70–2A7F	–	–	–	–	⩴	⩵	⩶	–	–	–	–	–	–	⩽	⩾	–
U+2A80–2A8F	–	–	–	–	–	⪅	⪆	–	–	⪇	⪈	–	–	–	–	–
U+2A90–2A9F	–	–	–	–	–	⪕	⪖	–	–	–	–	–	–	–	–	–
U+2AD0–2ADF	–	–	–	–	–	–	–	–	–	–	–	–	–	–	⫫	⫬
U+2AE0–2AEF	⫮	–	–	–	–	–	–	–	–	–	⫿	–	–	–	–	–
Private Use Area																
U+E000–E00F	_	_	–	–	–	–	–	–	–	–	–	–	–	–	–	–

Table 2: PennstanderMath-Regular *cont.*

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
U+E110-E11F	⊿	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Total number of glyphs shown from PennstanderMath-Regular: 748

5 Alphabets

5.1 Text version

– Normal/**Bold**/*Italic*/***BoldItalic***/*Slanted*/***BoldSlanted***:

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789

Slanted version are ***FakeSlanted*** *with 0.19 coefficient.*

5.2 Math version

– Normal/**Bold**/*Italic*/***BoldItalic***:

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789

– mathcal/mathfrak/mathbb:

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ

ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789